

1. Executive Summary

- The strategic shift to dual cycle development from single cycle development in 2025 has shown:
 - Effectively reduced risk of innovation, allowing the team to continue developing without interfering with the competition platform
 - Known reliability on the competition platform, with easier troubleshooting on known systems
 - Significantly reduced workload and pressure throughout the year on the competition platform
 - Improved validation of competition systems and methods with more testing time
 - Improved documentation due to the reduced workload
 - Better performance at competition, allowing for more preparation
- Competition and Parallel Cycles as of 2025:
 - NU25 is the fastest, most reliable and most tested NU Racing vehicle and should be continued to be tested in the first half of 2026. Two known issues should be resolved as well
 - NU26 will adopt new pouch cell accumulator, prompting changes to the chassis and Powerbox to improve centre of gravity and weight. Modular suspension is to be tested, allowing for both direct and inboard suspension, pending manufacturing and budget capabilities in 2026.
 - Compliance with FSAE-A needs to be verified before components ordered
 - The Autonomous platform needs as much support as NU Racing can give it, both team member and budget, it is highly likely that FSAE-A will start an AV division of competition in 2027
 - Monocoque development should be continued, although progress is slow
 - Hub Motor development should be prioritised given the performances of team at FSAE-A 2025
- If continued, future outcomes could be:
 - Consistent top 5 Overall Performances at FSAE-A
 - Reliable track days allowing for design validation and testing data to be collected
 - Consistent and improved NU Racing team member experience
 - More low risk innovation within NU Racing
 - Consistent improvement to NU Racing's competition platforms
 - Higher adaptability to outside challenges

2. Current Year Development Metrics (Cycle 1)

Metric	Target	Actual	Notes
NU25 Competition Compliant	100% by 1 st of Dec	98% by 1 st of Dec	Small fixes required at competition but first team through Technical Inspections
Issues Carried into Next Year	0 issues	2 issues	Motor tuning and SOC tuning, both known and documented with solutions.
On-Track Test Hours Logged	12 hr	16 hr 7 min	Various different driving styles across 16 track days, prioritising Autocross
Design Changes Validated	9 Completed	11 Completed	Page 7 of Management Design event booklet has all the validated systems.

- **Percentage of Track Day Testing Events Completed:** 73.9% (34/46 Completed)
- **Kilometres Completed:** 523.3 km (842.4km total for 2025)
- **New Lessons Learnt Logged:** 16 (Lessons are still yet to be logged by some team members)
- **New Documentation Created/Updated:** 11 (Across design, admin, and knowledge transfer)

3. Future Years Development Progress (Cycle 2)

Subsystem / Area	Progress Status	Notes
First Autonomous Drive to a Global Point	100%	Completed straight line, curved, object avoidance and stopping before an object.
Accumulator Design	90%	Cells are certified by UON, Mechanical Design is complete, Electrical schematics are complete and validated, PCB layouts need to be completed.
CFRP Monocoque Test Panel Compliance	100%	Using support from RPC, test panels were made, passing the requirements with a safety factor of 5
Hub Motor Gearbox Prepared for Prototype	100%	Research and design selection completed, 3D printed models made and tested.

- **Budget Percentage of Future Development Project:** 10.8%
- **Percentage of Team Working on Future Development Project:** 20% (6/25 including Leaders)

4. Comparison to July 2024 – Dec 2024

Metric	2024	2025	Notes
Competition Preparation	90% 1 Week from Comp	98% 1 Week from Comp	2024: NU24 was not operational, compliant, or reliable, design booklets, cost amendment and scenario incomplete, no comp plan and no comp sim. 2025: NU25 operational and 95% compliant and reliable, design booklets, cost amendment and scenario complete, Comp plan complete and Comp sim planned
On-Track Test Logged	372.2 km, 11 hr 12 min	523.3 km, 16 hr 07 min	11 track days vs 16 track days
Competition Performance	4 th place with no Awards	3 rd place with 6 Awards	Awards: Technical Inspection Award, 2 nd in Presentation, 2 nd in Acceleration, 1 st in Autocross, 3 rd in Efficiency, 3 rd Overall.
Design of Next Year’s Car	Mechanical: 50% Mechatronic: 30%	Mechanical: 75% Mechatronic: 60%	2024: Chassis, Accumulator done but missing compliance. Components to carry over was select. 2025: New Chassis, Accumulator, Powerbox, Suspension 90%+ complete, carried over components selected.

2024:

- **Very little documentation and knowledge transfer**
 - **Little to no design methodology documentation (only PCB DDR and ZTH)**
 - **Both Mechanical and Mechatronic Design had little to no validation**
 - **Unsolved issues from 2023**
 - **Larger budget and Capex yet only some small support for AVONE**
 - **AMS fault in Autocross and no wet tyres for Skidpad costing the team 3rd place at Competition**
 - **Master switch and front wheel bearing failures at competition as well as larger problems in tech (brake test and accumulator fuses)**
 - **Rushed testing period due to motor controller issues stopping driving for 2.5 months**
 - **Smaller team with less time to prepare for static events and finish car**
 - **Several unknown issues carried into the next year**
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2025:

- **Large Improvement to documentation and knowledge transfer**
 - **More design methodology documentation (How Tos, videos, DDR improvements, ZTH)**
 - **Both Mechanical and Mechatronic Design Changes validated**
 - **Solved issues carried from 2024**
 - **25% Budget cut but still supported parallel projects as well as competition**
 - **Maximised Competition output, collecting clear feedback for improvement in all events and achieved best NU Racing Overall finish and most awards ever**
 - **Small inverter issues during Acceleration but did not cost the team performance or result. Issue is known and a solution is known just not implemented due to time restraints**
 - **First team through technical inspections**
 - **In depth testing period, optimising vehicle dynamics as well as testing all systems and NU25 reliability**
 - **Reliability issues identified, requiring accumulator maintenance, resolved for competition**
 - **Larger team allowed more time to prepare for static events and technical inspections**
 - **Two known issues carried into the next year with methods to resolve the issues**
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5. Impact Metrics

Impact Metric	Outcome	Interpretation / Benefit
2026 Vehicle Design Completed by end of 2025	Mechanical: 75% Mechatronic: 60%	Mechanical Manufacture can start early and Mechatronic PCB design
Average Driving Per Track Day	32.7 km and 60 min (2024 – 33.8 km and 61 min)	2024 focused on Endurance due to major reliability issues while 2025 focused on Autocross thanks to a 42% increase of testing compared to 2024.
Average Team Workload and Pressure (Load/5 + Pressure/5 = /10, Lower is better)	Cycle 1 Team: 5.83/10 Cycle 2 Team: 7.5/10	Cycle 1 workload has decreased from 7.71 in previous years making the run to competition much easier correlating directly to a better performance.
Team Technical Satisfaction Score (Average of interest, enjoyability, learning and excitement out of 5, Higher is better)	Cycle 1 Team: 4.37/5 Cycle 2 Team: 4.58/5	Both Teams feel similar satisfaction, portraying that NU Racing delivers an excellent experience and matches or exceeds 4.38/5 from previous years

6. Risks and Mitigations

Identified Risk	Mitigation Strategy
Team growth exceeds budget size and number of scopes	Interview and only offer projects when it suits both NU Racing and the team member, critical decision making by leadership to ensure team is not overloaded, running projects thin
Future-Year Budget Overhead	NU Teams IPs can offer additional support, current year saving reallocated, and/or extra allocation in following year
Rushing to Finished Product	Ensure goal is responsible and regularly reconsidered, ensuring critical steps are taken and clear objectives are met, especially for parallel projects

7. Next Steps

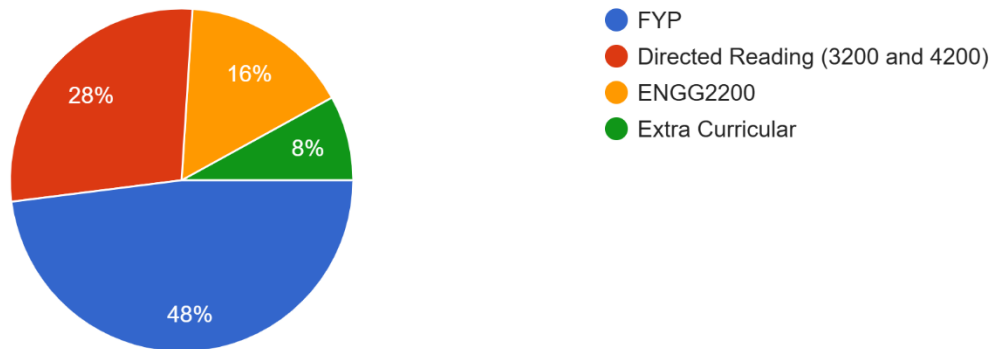
- Repeat feedback form in 2026, continuing the same development approach, validating methodology
- Fix the two issues being carried over to 2026 ASAP
- NU26 design and manufacture alongside NU25 testing
- Increase AVONE team, helping push towards a platform which can be taken to competition
- Continue validating new designs in 2026
- Continue support for Hub Motors and CFRP Monocoque
- Validate and extensively test the new pouch cell accumulator, utilizing the dyno

8. Appendices

- Data can be found in MoTeC GitHub and 2025 Track Day Testing Log
- Results of Feedback form excluding Long Responses:

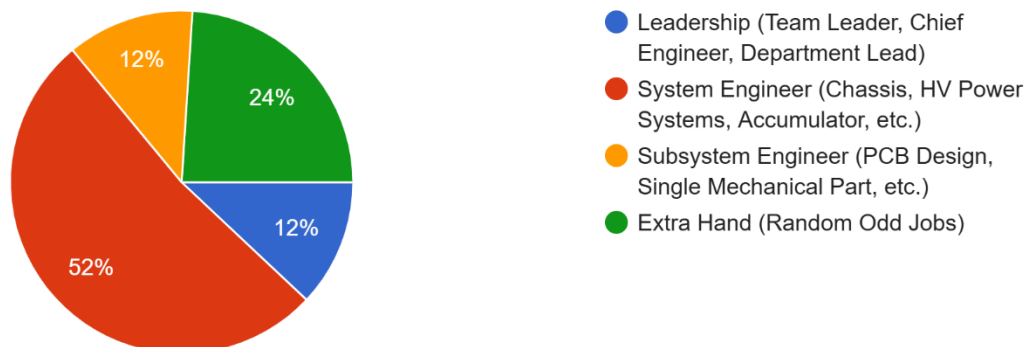
What was your level in the Team during Semester 2 2025?

25 responses



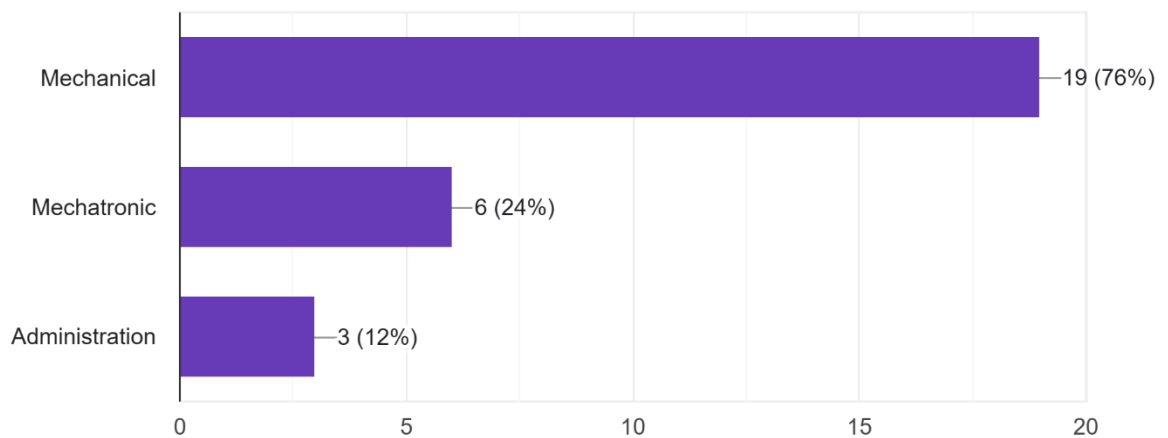
What was your Role in the Team during Semester 2 2025?

25 responses



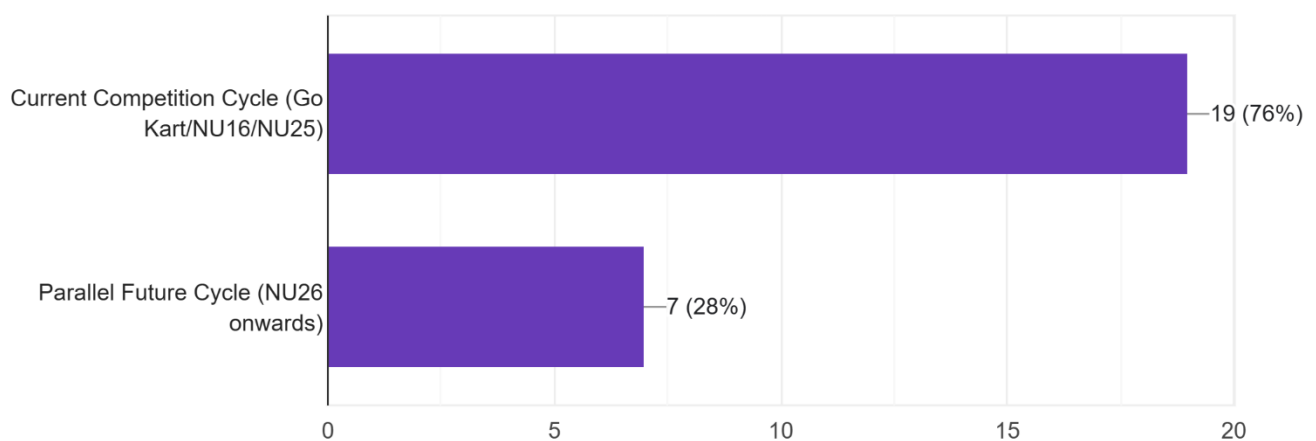
What Department did you work apart of primarily?

25 responses



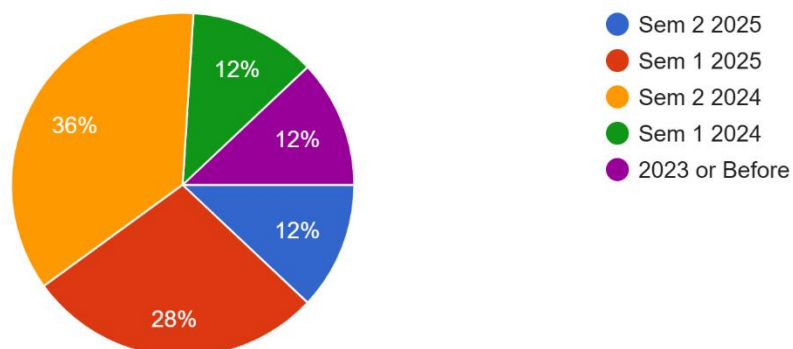
What design cycle have you worked on Primarily?

25 responses



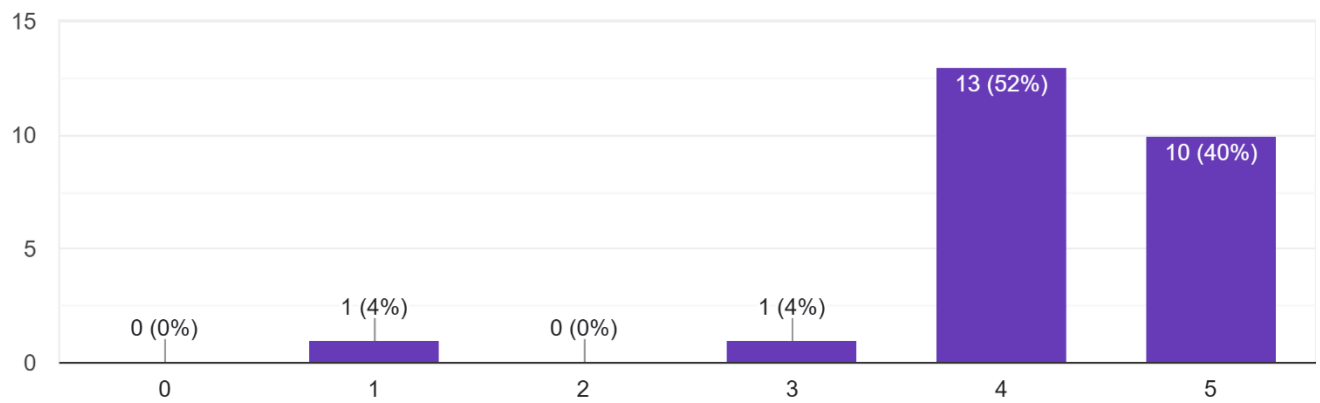
How long have you been in NU Racing at any Level?

25 responses



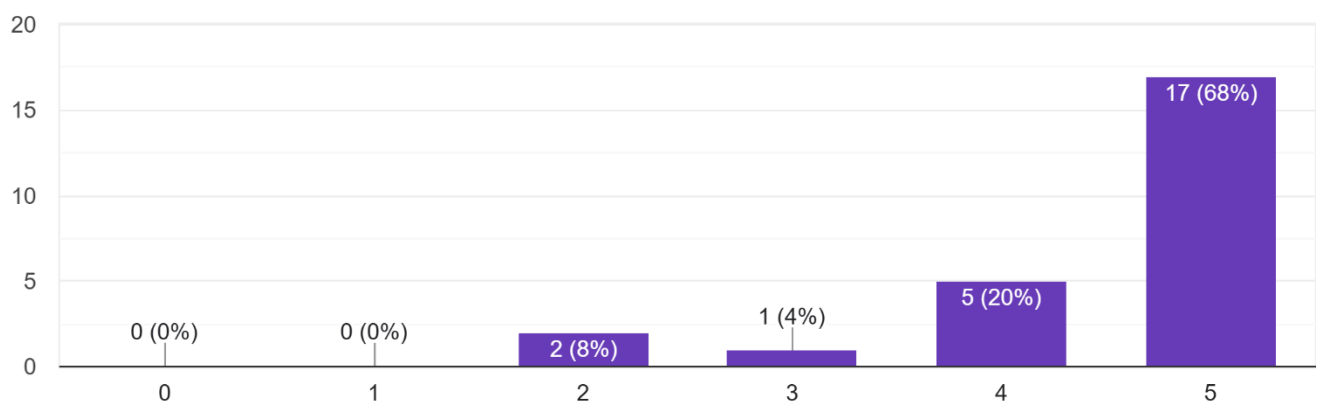
How much have you enjoyed the work you have done?

25 responses



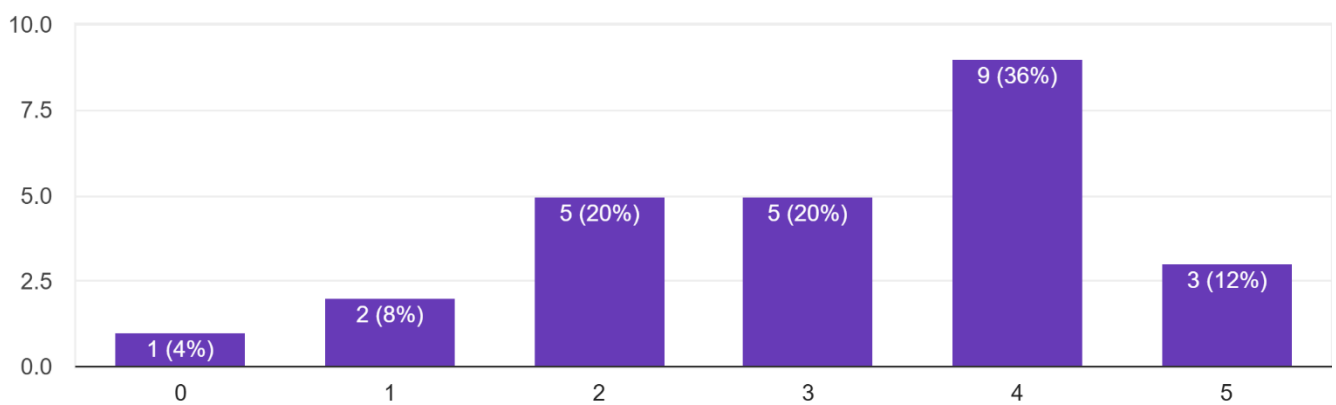
How interested have you been in the work you have completed?

25 responses



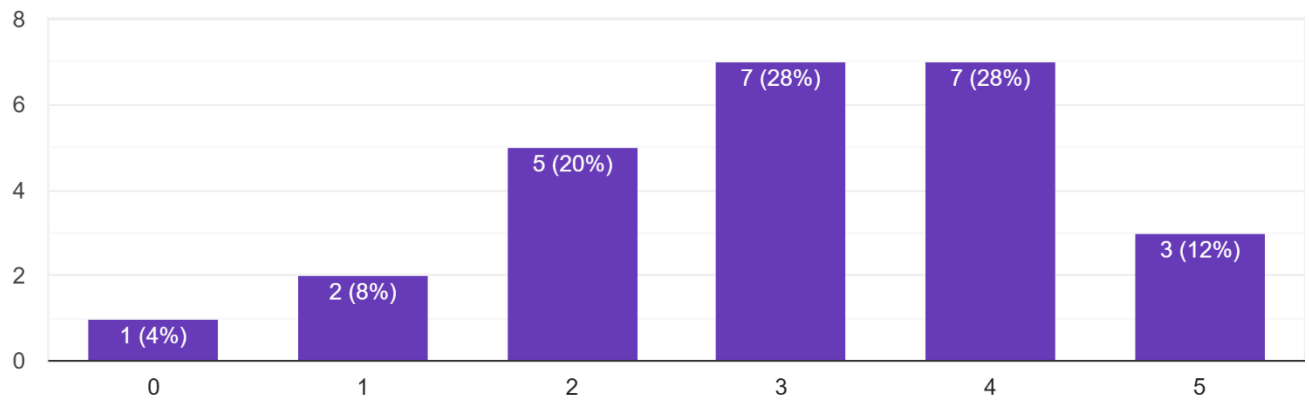
How manageable was your work load?

25 responses



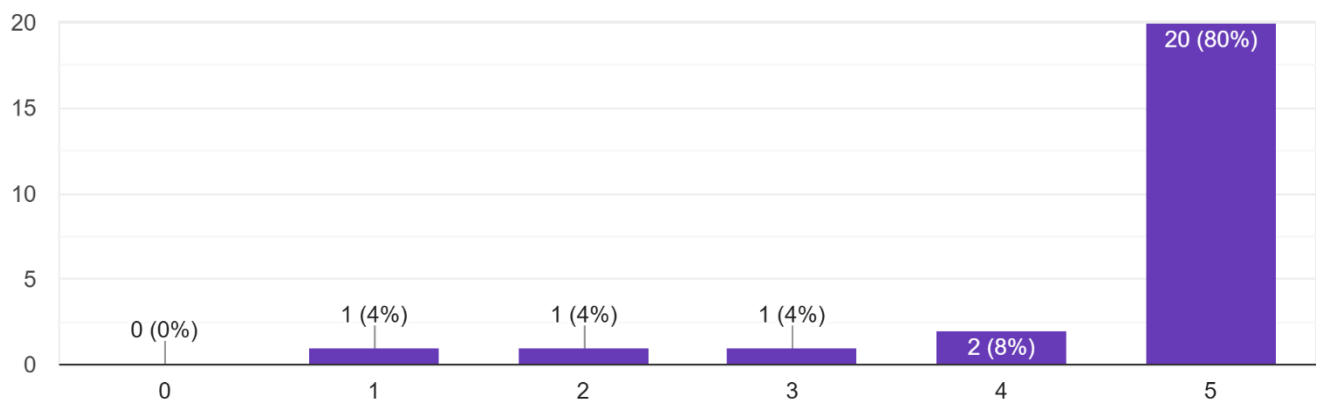
How much pressure did you feel?

25 responses



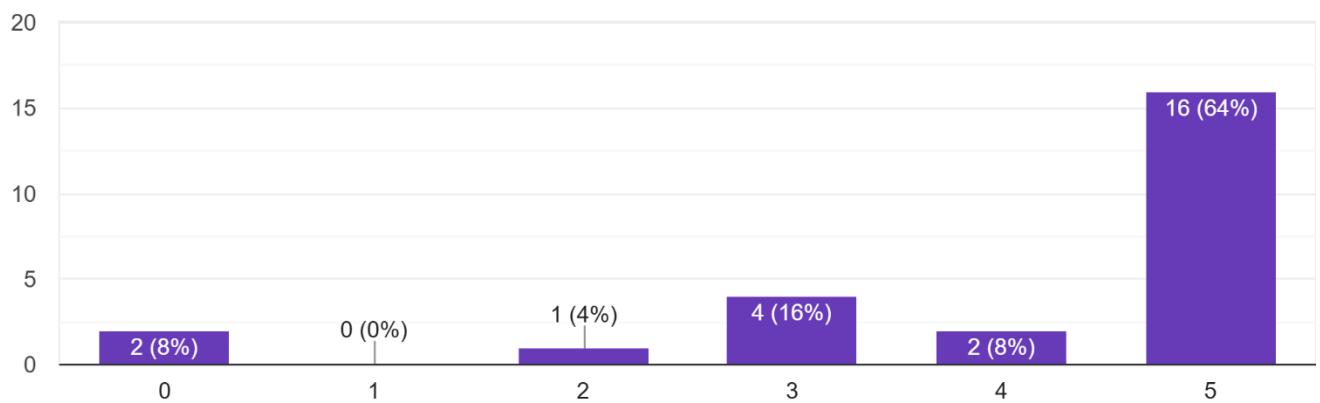
Do you feel you have increased your engineering understanding?

25 responses



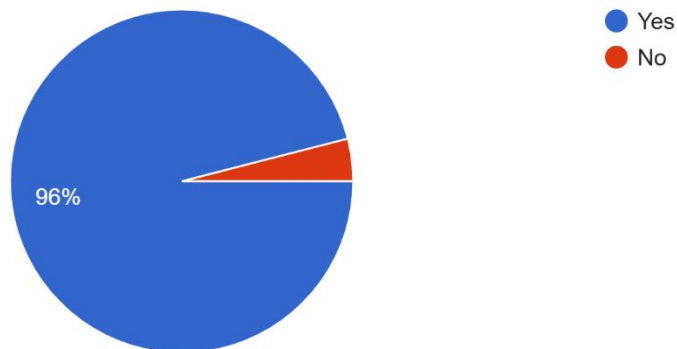
Are you excited to continue with NU Racing

25 responses



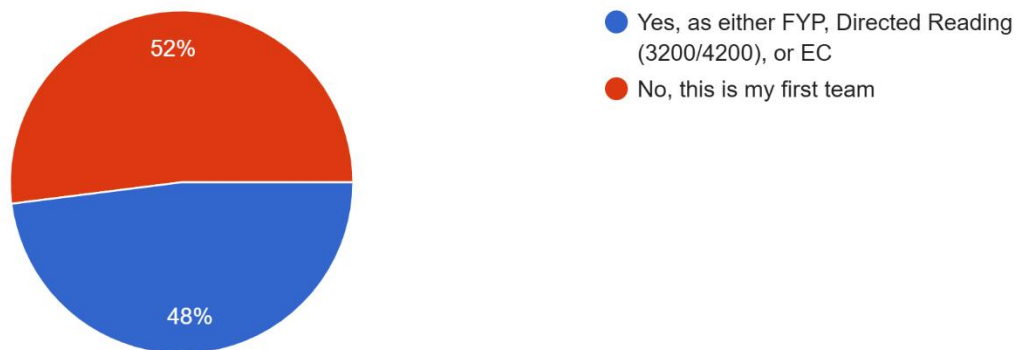
Do you like how the team does technical development?

25 responses



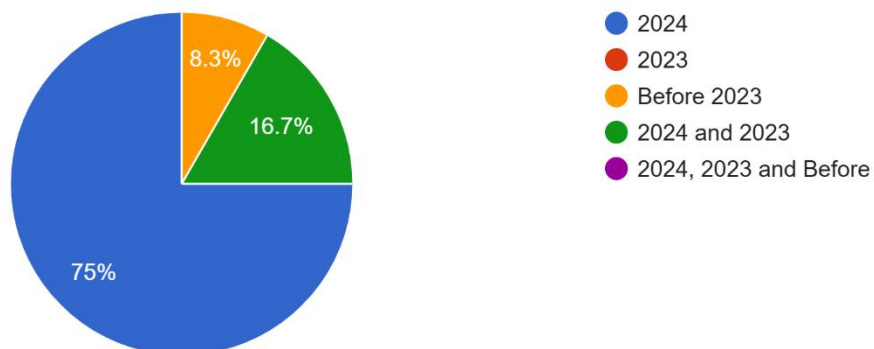
Have you been apart of a previous NU Racing team?

25 responses



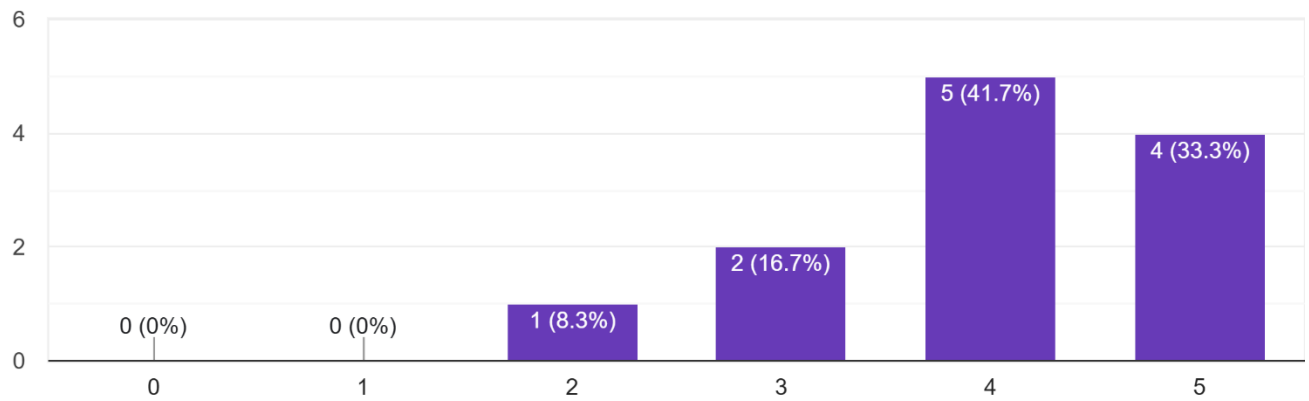
What year were you involved with?

12 responses



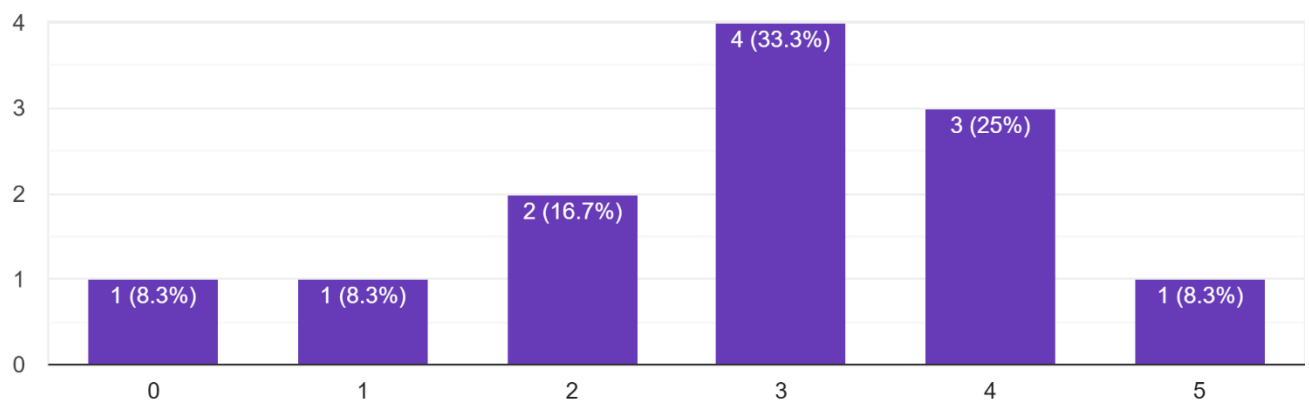
How much have you enjoyed the work you have done during that year/those years?

12 responses



How manageable was your work load during that year/those years?

12 responses



How much pressure did you feel during that year/those years?

12 responses

